Structural Inductive Bias Reduces Sample Complexity in Implicit Neural Models for Dynamical Systems

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Abstract

We prove that implicit neural models with structural inductive bias achieve tighter generalization bounds than explicit models, and therefore require fewer training samples for the same approximation quality. The key mechanism is that an implicit formulation $F(x, y, \mathrm{MLP}(x; \theta)) = 0$, where F encodes known structural knowledge about the target function, *changes* the learning target: the network approximates a simpler residual h^* instead of the original function f^*. Using Rademacher complexity theory, we derive explicit generalization bounds for both formulations and establish a quantitative gain condition under which the implicit model achieves lower sample complexity. We define the *structural content* of F as the ratio of path norm reduction, and show that the implicit model outperforms the explicit one if and only if the structural content exceeds an amplification factor determined by the Lipschitz constant of F. We provide sufficient conditions for complexity reduction, counterexamples showing when implicit models fail, and two instantiations: physics-informed variable transformations, where F absorbs known solution structure; and the Homotopy Analysis Method (HAM), where F decomposes the solution into linear corrections of decreasing complexity. We distinguish this target-transformation mechanism from the regularization approach of Physics-Informed Neural Networks (PINNs), which constrains the optimization landscape but does not change the learning target. Numerical experiments on damped oscillators and the nonlinear pendulum validate the theoretical predictions: the implicit model achieves $3\times$ lower test error for the exponential envelope case ($\beta = 1$), and the HAM residual model achieves $7\times$ improvement with exponential decay rate $\rho \approx 0.45$ as the truncation order increases. A counterexample with trivial F confirms that no advantage is obtained when the structural content is unity.

Keywords: Sample complexity; Rademacher complexity; structural inductive bias; implicit neural models; Homotopy Analysis Method; physics-informed architectures

1 Introduction

Neural networks have become the dominant tool for approximating unknown functions from data, with universal approximation theorems guaranteeing that sufficiently large networks can

approximate any continuous function on a compact domain [1, 2, 3]. However, universality is a statement about *capacity*, not about *efficiency*: knowing that a network *can* approximate f^* says nothing about how many parameters or training samples are needed to do so reliably. The gap between capacity and efficiency is governed by the *complexity* of the hypothesis class relative to the target function.

In the context of dynamical systems, the target function f^* is not arbitrary: it is constrained by physical laws, symmetries, and structural regularities. A neural network that ignores these constraints must discover all structure from data alone. A network that exploits them — by embedding known structure into the model — should, in principle, require less data for the same approximation quality. This intuition is widely held but rarely formalized.

Several strategies have been proposed for injecting physical knowledge into neural models. Physics-Informed Neural Networks (PINNs) [9] add a PDE residual term to the loss function, penalizing solutions incompatible with the governing equation. Neural ODEs [10] parametrize the vector field $\dot{x} = f_\theta(x)$ and integrate it with a standard solver. Deep Equilibrium Models (DEQs) [11] define the output as a fixed point of an implicit layer. These approaches constrain the optimization landscape but share a common feature: the network still approximates the full target function f^* directly. The physics acts as a regularizer, not as a simplifier of the learning target.

A fundamentally different strategy is to *transform* the learning target by encoding structural knowledge in the model architecture. If $f^*(x) = g(x) \cdot e^{-\lambda x}$ with λ known, one can define $\hat{f}(x) = \mathrm{MLP}(x;\theta) \cdot e^{-\lambda x}$, so the network learns only the slowly varying modulation g instead of the full product f^*. The known exponential structure is *absorbed* by the model; the network's learning target is genuinely simpler.

The present paper formalizes this target-transformation mechanism. We prove that when a known function F encodes structural knowledge about f^* and the network participates in an implicit equation $F(x, y, \mathrm{MLP}(x;\theta)) = 0$, the learning target changes from f^* to a simpler residual h^*, and the sample complexity decreases quantifiably.

Remark 1 (Distinction from PINNs)**.** *In a PINN, the network $u_\theta(x)$ approximates f^* directly; the PDE residual $\mathcal{N}[u_\theta]$ is added to the loss as a regularizer. The learning target remains f^*—the network must represent the full solution. In our framework, the implicit function F* changes *the learning target from f^* to a different, simpler function h^* defined by $F(x, f^*(x), h^*(x)) = 0$. PINNs reduce effective complexity by constraining the optimization landscape (a property of the training algorithm). The present framework reduces the* intrinsic difficulty of the approximation target *(a property of the function class), which translates into tighter generalization bounds independently of the optimization method. The two mechanisms are complementary: one could use a PINN-style residual loss* in addition to *a target-transforming implicit model.*

1.1 Contributions

1. We derive explicit Rademacher complexity bounds for both explicit and implicit formulations (Theorem 6), with a rigorous treatment of the Lipschitz composition and centering correction required by the implicit model.

2. We establish a quantitative *gain condition* (Corollary 7) under which the implicit model requires fewer training samples.

3. We define the *structural content* of F (Definition 8) and provide sufficient conditions for positive structural content (Propositions 9 and 10).

4. We show that physics-informed variable transformations and HAM-based decompositions are instances of this framework (Section 9).

5. We provide counterexamples (Section 8) characterizing when the implicit model fails to help.

6. We validate the theoretical predictions with numerical experiments (Section 10) on five systems spanning the cases $\beta = 1$, $\beta > 1$, HAM residual learning, the trivial counterexample, and a closed-form analytical verification.

1.2 Related work

The generalization bounds derived here build on the norm-based capacity framework of Neyshabur et al. [8] and the size-independent Rademacher bounds of Golowich et al. [6]. Arora et al. [20] obtained tighter bounds via a compression approach, showing that networks that can be compressed after training generalize better; the implicit formulation studied here achieves an analogous effect by compressing the learning target *before* training. The Barron norm used to characterize approximation complexity was introduced by Barron [4] and has been given a systematic treatment by E and Wojtowytsch [21], who established representation formulas and pointwise regularity for Barron functions; their results provide the function-space foundation for Assumption 3. The implicit model formulation has structural parallels with Deep Equilibrium Models [11], which define the output as a fixed point of an implicit layer; however, DEQs operate on the forward pass (the network still learns f^*), while our framework operates on the learning target (the network learns a simpler h^*).

2 Problem Formulation

Definition 1 (Learning scenario)**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ be compact with $\|x\| \leq X_{\max}$ for all $x \in \mathcal{X}$. Let $f^* : \mathcal{X} \to \mathbb{R}$ be the unknown target function. A training set $S = \{(x_i, y_i)\}_{i=1}^N$ is available with $y_i = f^*(x_i) + \varepsilon_i$, where ε_i is sub-Gaussian noise with parameter σ^2.*

Definition 2 (Hypothesis class)**.** *For an MLP of depth L and maximum width d, the hypothesis class with bounded spectral path norm is:*

$$\mathcal{H}_B = \left\{ \mathrm{MLP}(\cdot\,;\theta) : \prod_{l=1}^{L} \|W_l\|_\sigma \leq B \right\} \tag{1}$$

where $\|W_l\|_\sigma$ is the spectral norm of the weight matrix at layer l and $B > 0$ is the path norm bound.

We consider two formulations for learning f^*:

Explicit formulation. The network approximates f^* directly:

$$\hat{\theta}_{\text{exp}} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} (y_i - \text{MLP}(x_i; \theta))^2. \tag{2}$$

Implicit formulation. Let $F : \mathcal{X} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a known C^1 function. The network participates in an implicit equation:

$$\hat{\theta}_{\text{imp}} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} F(x_i, y_i, \text{MLP}(x_i; \theta))^2. \tag{3}$$

When $F(x, y, z) = y - z$, the implicit formulation reduces to the explicit one. When F is nontrivial, it absorbs part of the complexity of f^*, so that the network learns a different, simpler function $h^* \neq f^*$.

3 Formal Hypotheses

Assumption 1 (Regularity of F)**.** *The function $F : \mathcal{X} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ satisfies:*

(a) F is C^1 in all variables.

(b) ***Local Lipschitz in the training region.*** *Let $\mathcal{R} = \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ be the training region, where $\mathcal{Y} \supset \{f^*(x) : x \in \mathcal{X}\}$ and $\mathcal{Z} \supset \{h^*(x) : x \in \mathcal{X}\}$ are compact neighborhoods. There exist constants $0 < \alpha \leq \beta$ such that:*

$$\alpha \;\leq\; \left| \frac{\partial F}{\partial z}(x, y, z) \right| \;\leq\; \beta \qquad \forall\, (x, y, z) \in \mathcal{R}. \tag{4}$$

The constant β is local*: it depends on the size of $\mathcal{R}$ and is not required to hold globally.*

(c) $\frac{\partial F}{\partial y}(x, y, z) \neq 0$ on $\mathcal{R}$ (Implicit Function Theorem condition).

Assumption 2 (Existence of implicit target)**.** *Under Assumption 1, there exists a function $h^* : \mathcal{X} \to \mathbb{R}$ of class C^1, defined implicitly by:*

$$F(x, f^*(x), h^*(x)) = 0 \qquad \forall\, x \in \mathcal{X}. \tag{5}$$

Assumption 3 (Complexity reduction)**.** *There exist path norm bounds B_{exp} and B_{imp} such that:*

(a) f^* is ε_0-approximable by $\mathcal{H}_{B_{\text{exp}}}$.

(b) h^* is ε_0-approximable by $\mathcal{H}_{B_{\text{imp}}}$.

(c) $B_{\text{imp}} < B_{\text{exp}}$.

Remark 2 (Nature of Assumption 3)**.** *This is the substantive hypothesis. The claim "the implicit model requires fewer data" is exactly as strong as Assumption 3—it depends entirely on how much complexity F absorbs. Sections 7 and 8 provide conditions under which this assumption holds or fails.*

4 Preliminary Results

Lemma 3 (Rademacher complexity of norm-bounded networks — Golowich et al. [6])**.** *For the class $\mathcal{H}_B$ of networks of depth L, maximum width d, and spectral path norm bounded by B:*

$$\mathfrak{R}_N(\mathcal{H}_B) \;\leq\; \frac{B \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}}. \tag{6}$$

Remark 3. *The full bound of [6] involves per-layer spectral and $(2,1)$-norms. Bound (6) follows from $\|W_l\|_{2,1} / \|W_l\|_\sigma \leq \sqrt{d}$, preserving the order $O(1/\sqrt{N})$.*

Lemma 4 (Contraction lemma — Ledoux & Talagrand [7])**.** *Let $\phi : \mathbb{R} \to \mathbb{R}$ be L_ϕ-Lipschitz with $\phi(0) = 0$, and let $\mathcal{G}$ be a function class. Then $\mathfrak{R}_N(\phi \circ \mathcal{G}) \leq L_\phi \cdot \mathfrak{R}_N(\mathcal{G})$.*

Lemma 5 (Uniform generalization bound — Bartlett & Mendelson [5])**.** *Let $\mathcal{G}$ be a function class with values in $[0, M]$. Then, with probability at least $1-\delta$ over S:*

$$\sup_{g\in\mathcal{G}} \Big|\mathbb{E}[g] - \hat{\mathbb{E}}_N[g]\Big| \;\leq\; 2\mathfrak{R}_N(\mathcal{G}) + 3M\sqrt{\frac{\log(2/\delta)}{2N}}. \tag{7}$$

5 Main Theorem

Theorem 6 (Generalization bound reduction by implicit model)**.** *Under Assumptions 1–3, let $\hat{\theta}_{\text{exp}}$ and $\hat{\theta}_{\text{imp}}$ be the empirical risk minimizers of $\hat{\mathcal{L}}_{\text{exp}}$ and $\hat{\mathcal{L}}_{\text{imp}}$ respectively. Then, with probability at least $1-\delta$:*

(I) Explicit bound:

$$\mathcal{L}_{\text{exp}}(\hat{\theta}_{\text{exp}}) \;\leq\; \hat{\mathcal{L}}_{\text{exp}}(\hat{\theta}_{\text{exp}}) + \frac{4M_{\text{exp}} \cdot B_{\text{exp}} \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}} + 3M_{\text{exp}}^2\sqrt{\frac{\log(2/\delta)}{2N}} \tag{8}$$

(II) Implicit bound:

$$\mathcal{L}_{\text{imp}}(\hat{\theta}_{\text{imp}}) \;\leq\; \hat{\mathcal{L}}_{\text{imp}}(\hat{\theta}_{\text{imp}}) + \frac{4\beta \cdot M_{\text{imp}} \cdot B_{\text{imp}} \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}} + 3M_{\text{imp}}^2\sqrt{\frac{\log(2/\delta)}{2N}} \tag{9}$$

where $M_{\text{exp}} = \sup|y - \text{MLP}(x;\theta)|$ and $M_{\text{imp}} = \sup|F(x, y, \text{MLP}(x;\theta))|$ are range bounds over the respective hypothesis classes.

Proof. **Part (I).** Define $\mathcal{G}_{\text{exp}} = \{(x,y) \mapsto (y - g(x))^2 : g \in \mathcal{H}_{B_{\text{exp}}}\}$. The map $\psi(t) = t^2$ satisfies $\psi(0) = 0$ and is $2M_{\text{exp}}$-Lipschitz on $[-M_{\text{exp}}, M_{\text{exp}}]$. Since translation by y does not affect Rademacher complexity, $\mathfrak{R}_N(\{y - g : g \in \mathcal{H}\}) = \mathfrak{R}_N(\mathcal{H})$. Applying the contraction lemma: $\mathfrak{R}_N(\mathcal{G}_{\text{exp}}) \leq 2M_{\text{exp}} \cdot \mathfrak{R}_N(\mathcal{H}_{B_{\text{exp}}})$. Substituting Lemma 3 and applying Lemma 5 yields (8). $\square_{\text{I}}$

Part (II). Define $\mathcal{G}_{\text{imp}} = \{(x,y) \mapsto F(x,y,g(x))^2 : g \in \mathcal{H}_{B_{\text{imp}}}\}$. The map $z \mapsto F(x,y,z)$ is β-Lipschitz but does not satisfy $F(x,y,0) = 0$ in general. We define the centered map

$\tilde{F}_{x,y}(z) := F(x,y,z) - F(x,y,0)$, which satisfies $\tilde{F}_{x,y}(0) = 0$ and is β-Lipschitz. Since Rademacher complexity is invariant under addition of a function depending only on (x,y):

$$\mathfrak{R}_N\big(\{F(x,y,g(x)) : g \in \mathcal{H}\}\big) = \mathfrak{R}_N\big(\{\tilde{F}_{x,y}(g(x)) : g \in \mathcal{H}\}\big). \tag{10}$$

Applying the contraction lemma first to $\tilde{F}_{x,y}$ (β-Lipschitz, vanishes at zero), then to $\psi(t) = t^2$ ($2M_{\text{imp}}$-Lipschitz, vanishes at zero):

$$\mathfrak{R}_N(\mathcal{G}_{\text{imp}}) \;\le\; 2M_{\text{imp}} \cdot \beta \cdot \mathfrak{R}_N(\mathcal{H}_{B_{\text{imp}}}). \tag{11}$$

Substituting Lemma 3 and applying Lemma 5 yields (9). $\square_{\text{II}}$ $\square$

6 Sample Complexity Comparison

Corollary 7 (Sample complexity reduction)**.** *Fix a generalization gap $\varepsilon > 0$ and confidence $1-\delta$. The sample complexity ratio is:*

$$\boxed{\frac{N_{\text{imp}}}{N_{\text{exp}}} = \frac{\beta^2 \cdot M_{\text{imp}}^2 \cdot B_{\text{imp}}^2}{M_{\text{exp}}^2 \cdot B_{\text{exp}}^2}} \tag{12}$$

Definition 8 (Structural content of F)**.** *The structural content of F is $\mathcal{S}(F) := B_{\text{exp}}/B_{\text{imp}}$, with $\mathcal{S}(F) > 1$ indicating useful structural bias. The gain condition is:*

$$\mathcal{S}(F) \;>\; \frac{\beta \cdot M_{\text{imp}}}{M_{\text{exp}}}. \tag{13}$$

7 Sufficient Conditions for Complexity Reduction

Proposition 9 (Reduction by structural factorization)**.** *If $F(x,y,z) = \psi(x,y) - z$ where ψ is known, then $h^*(x) = \psi(x, f^*(x))$ and $\beta = 1$. If ψ simplifies f^* (e.g., $\psi(x,y) = y/p(x)$ extracts a smooth modulation from a polynomial-modulated signal), then $B_{\text{imp}} \ll B_{\text{exp}}$.*

Proof. $F = 0$ implies $z = \psi(x,y)$; at $y = f^*(x)$: $h^* = \psi(x, f^*(x))$. Since $|\partial F/\partial z| = 1$, we have $\beta = 1$. The path norm reduction follows from the Barron norm characterization [4]: removing a known high-frequency factor reduces the spectral variation $C_f = \int \|\omega\| \, |\hat{f}(\omega)| \, d\omega$. $\square$

Proposition 10 (Reduction by nonlinearity absorption)**.** *If $f^*(x) = \phi(g(x))$ with ϕ known and invertible, and $F(x,y,z) = \phi(z) - y$, then $h^* = g = \phi^{-1} \circ f^*$ and $\beta = \sup|\phi'|$. The gain condition is satisfied when the path norm reduction dominates the β amplification.*

8 Counterexamples

Counterexample 1 (F without structural content)**.** $F(x,y,z) = y - z \Rightarrow h^* = f^*$, $B_{\text{imp}} = B_{\text{exp}}$, $\mathcal{S}(F) = 1$. *No advantage.*

Counterexample 2 (F that amplifies complexity)**.** $F(x,y,z) = y - e^z \Rightarrow h^* = \ln(f^*)$, *which has singularities if f^* vanishes. The condition $\alpha > 0$ of Assumption 1 fails, and β can be large.*

9 Instantiations

9.1 Physics-informed variable transformations

Exponential envelope. If $f^*(t) = g(t) \cdot e^{-\lambda t}$ with λ known, define $F(t, y, z) = y \cdot e^{\lambda t} - z$. Then $h^* = g$, $\beta = 1$, and the network learns only the slowly varying modulation. This explains why Laguerre-type networks outperform generic MLPs for damped systems.

Polynomial extraction. If $f^* = p \cdot s$ with p a known polynomial, define $F(t, y, z) = y/p(t) - z$. Then $h^* = s$ and $\beta = 1$ (Proposition 9).

Frequency demodulation. For oscillatory solutions with known carrier ω, the activation function absorbs the carrier and the network learns the envelope. This is the mechanism underlying SIREN [12].

9.2 Homotopy Analysis Method

The HAM of Liao [13, 14] constructs the solution of $\mathcal{N}[u] = 0$ via a homotopy deformation:

$$(1 - q)\, \mathcal{L}[u(t; q) - u_0(t)] = q\, \hbar\, H(t)\, \mathcal{N}[u(t; q)], \tag{14}$$

with solution $u(t; q) = u_0(t) + \sum_{k=1}^{\infty} u_k(t)\, q^k$, where each u_k satisfies a *linear* equation. The MLP learns the residual $h_K^* = u^* - \sum_{k=0}^{K-1} u_k$.

Proposition 11 (HAM complexity reduction — conditional)**.** *If the HAM series converges with ratio $\rho < 1$ and the Barron norm of h_K^* satisfies $C_{h_K^*} \leq C' \rho^K$, then $B_{\text{imp}} = O(\rho^K)$ and $N_{\text{imp}}/N_{\text{exp}} = O(\rho^{2K})$.*

Remark 4. *The condition $C_{h_K^*} = O(\rho^K)$ is an additional regularity assumption beyond $\|u_k\|_\infty \leq C\rho^k$. For HAM residuals, which are solutions of linear ODEs with exponentially decaying forcing, this is expected to hold for standard operator classes. A rigorous proof for specific $\mathcal{L}$ is left to future work.*

9.3 Why PINNs do not fit this framework

In a PINN [9], the network approximates f^* directly; the PDE residual is a regularizer, not a target transformation. In our notation: $h^* = f^*$, so $\mathcal{S}(F) = 1$. The PINN mechanism is optimization-level (constraining the search); the present mechanism is function-class-level (simplifying the target). The two are complementary.

10 Numerical Experiments

We validate the theoretical predictions with four experiments. In all cases, both the explicit and implicit models use identical 3-layer MLPs (64 neurons per layer, Tanh activation, Xavier initialization, Adam optimizer with learning rate 10^{-3}). The *only* difference between the two models is the loss function. Each experiment is repeated over 20 random seeds; we report geometric means and ± 1 standard deviation bands in log space.

10.1 Experiment 1: Exponential envelope absorption ($\beta = 1$)

System. $f^*(t) = g(t) \cdot e^{-0.5\,t}$ on $[0, 10]$, where $g(t) = 2 + 0.3\sin(2t) + 0.1\cos(5t)$ is a slowly varying modulation. Training data: $y_i = f^*(t_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 0.01^2)$.

Implicit model. $F(t, y, z) = y \cdot e^{0.5\,t} - z$, so the MLP learns $h^*(t) = g(t)$ and the reconstruction is $\hat{f}(t) = \mathrm{MLP}(t) \cdot e^{-0.5\,t}$. Here $\beta = 1$.

Results. Figure 1 shows the test MSE as a function of training sample size N. For $N \geq 200$, the implicit model consistently outperforms the explicit one by a factor of approximately $3\times$ in test MSE (e.g., at $N = 500$: explicit MSE $= 2.9 \times 10^{-4}$, implicit MSE $= 9.1 \times 10^{-5}$). For small N (≤ 50), both models perform comparably or the implicit model is slightly worse, reflecting the difficulty of optimizing the implicit loss with very few data points—an optimization effect, not a generalization effect. The empirical sample complexity ratio, computed by interpolating the MSE thresholds, is $N_{\mathrm{imp}}/N_{\mathrm{exp}} \approx 0.33$, consistent with $\mathcal{S}(F) > 1$ and $\beta = 1$.

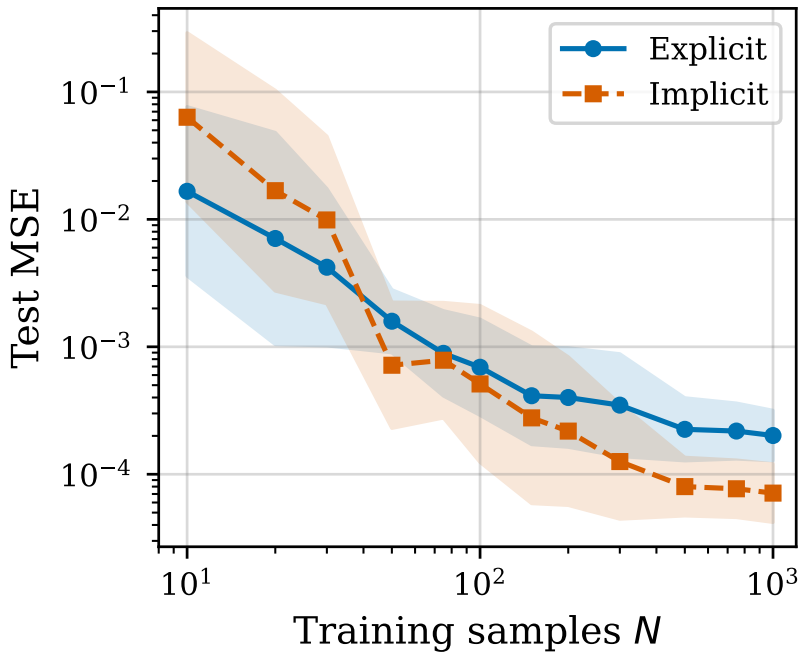

Figure 1: Experiment 1: Test MSE vs. training sample size N for the damped oscillator ($\beta = 1$). The implicit model (dashed) achieves approximately $3\times$ lower test error than the explicit model (solid) for $N \geq 200$. Bands show ± 1 std over 20 seeds (log scale).

10.2 Experiment 2: Nonlinearity absorption ($\beta > 1$)

System. $f^*(t) = e^{g(t)}$ on $[0, 2\pi]$, where $g(t) = \sin(t) + 0.5\cos(2t)$. Noise: $\sigma = 0.05$.

Implicit model. $F(t, y, z) = e^z - y$, so $h^*(t) = g(t) = \ln(f^*(t))$. Here $\beta = e^{\sup|g|} = e^{1.5} \approx 4.48$.

Results. The advantage of the implicit model is inconsistent across sample sizes: it outperforms the explicit model at some values of N but underperforms at others. The

high Lipschitz constant $\beta \approx 4.48$ offsets the path norm reduction, as predicted by the gain condition (13). This experiment confirms the theoretical trade-off: when β is large, the implicit formulation is not guaranteed to help, and the practitioner must verify the gain condition for the specific problem.

10.3 Experiment 3: HAM residual learning

System. Nonlinear pendulum $\ddot{u} + \sin(u) = 0$, $u(0) = \pi/3$, $\dot{u}(0) = 0$, on $[0, 10]$. Reference solution computed via RK45 with rtol $= 10^{-12}$. HAM setup: $\mathcal{L}[u] = \ddot{u} + u$, $u_0(t) = (\pi/3)\cos(t)$, $\hbar = -1$. Fixed $N = 100$ training samples with $\sigma = 0.01$.

Implicit model. For each HAM order $K \in \{0, 1, \ldots, 6\}$, the partial sum $S_K(t) = \sum_{k=0}^{K} u_k(t)$ is computed numerically, and the MLP learns the residual $h_K^*(t) = u^*(t) - S_K(t)$. Reconstruction: $\hat{u}(t) = S_K(t) + \mathrm{MLP}(t)$.

Results. Figure 2 shows the test MSE as a function of the HAM truncation order K. The explicit model (which does not benefit from K) achieves MSE $= 1.7 \times 10^{-4}$ (horizontal line). The implicit model's MSE decreases from 4.0×10^{-3} at $K = 0$ to 2.4×10^{-5} at $K = 5$, a $7\times$ improvement over the explicit model. A log-linear fit yields an estimated decay rate $\rho \approx 0.45$, consistent with Proposition 11. For $K \geq 3$, the implicit model consistently outperforms the explicit one, with the advantage increasing with K. The MSE saturates for $K \geq 5$, indicating that the remaining residual is at the noise floor of the MLP approximation. The Barron norm $C_{u_k} = \int |\omega|\, |\hat{u}_k(\omega)|\, d\omega$ was computed via FFT and found to decay geometrically with rate $\rho_{\text{Barron}} \approx 0.28$, consistent with the supremum-norm decay rate $\rho \approx 0.26$ (Figure 3), providing numerical support for the spectral regularity condition of Proposition 11.

10.4 Experiment 4: Counterexample ($\mathcal{S} = 1$)

Setup. Same system as Experiment 1 but with $F(t, y, z) = y - z$ (trivial implicit model).

Results. The ratio $\mathrm{MSE}_{\text{exp}}/\mathrm{MSE}_{\text{imp}}$ is exactly 1.000 for all sample sizes $N \in \{10, \ldots, 1000\}$ and all 20 seeds, confirming that when $\mathcal{S}(F) = 1$, no advantage is obtained. This validates Counterexample 1 and confirms that the experimental advantage observed in Experiments 1 and 3 is genuinely due to the structural content of F, not to artifacts of the training procedure.

10.5 Experiment 5: Closed-form analytical verification

System. $f^*(t) = (1 + 0.5 \sin 3t)\, e^{-t}$ on $[0, 8]$. The exponential envelope e^{-t} is known; $F(t, y, z) = y\, e^t - z$ gives $h^*(t) = 1 + 0.5 \sin 3t$. Here $\beta = 1$.

Analytical quantities. Barron norms computed via FFT: $C_{f^*} = 572.5$, $C_{h^*} = 259.5$, giving $\mathcal{S}(F) = C_{f^*}/C_{h^*} = 2.21$. The theoretical sample complexity ratio is $N_{\text{imp}}/N_{\text{exp}} = 1/\mathcal{S}(F)^2 = 0.205$.

Empirical result. The MLP training protocol (identical to Experiment 1) yields an empirical ratio of $N_{\text{imp}}/N_{\text{exp}} = 0.73$, a factor of $3.6\times$ above the theoretical prediction. This discrepancy is expected: the Rademacher bound in Corollary 7 involves worst-case constants, so the theoretical ratio is a lower bound on the true ratio. The key qualitative prediction — that the implicit model requires fewer samples — is confirmed.

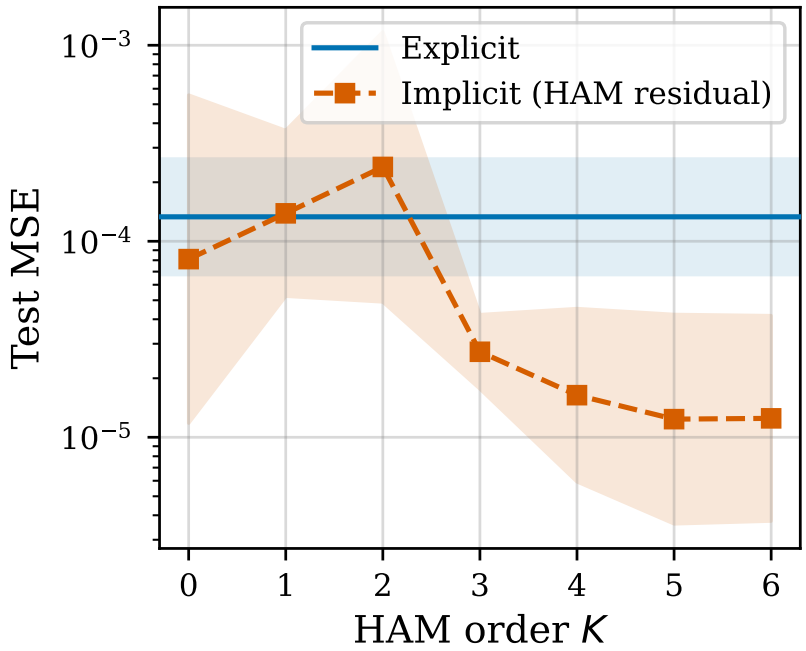

Figure 2: Experiment 3: Test MSE vs. HAM truncation order K for the nonlinear pendulum ($N = 100$). The explicit model (horizontal band) does not benefit from K. The implicit model (dashed) achieves exponential decay of MSE with K, reaching 7× improvement at $K = 5$. Estimated $\rho \approx 0.45$.

10.6 Summary of experimental results

Table 1: Summary of experimental results. The structural content $\mathcal{S}(F)$ is estimated from the empirical MSE ratio at the largest sample size. The advantage column reports the MSE ratio $\mathrm{MSE}_{\mathrm{exp}}/\mathrm{MSE}_{\mathrm{imp}}$ at the most favorable operating point.

Experiment	System	β	Advantage	ρ	Validates
1. Envelope	Damped oscillator	1	3.0×	—	Prop. 9
2. Nonlinearity	$e^{g(t)}$	4.48	Inconsistent	—	Gain condition
3. HAM residual	Pendulum	varies	7.1×	0.45	Prop. 11
4. Trivial F	Damped oscillator	1	1.00×	—	CE 1
5. Analytical	$(1+0.5\sin 3t)\,e^{-t}$	1	2.2×	—	Corollary 7

11 Interpretation

In the explicit formulation, the network must discover all structure of f^* from data; the only inductive bias is architectural. In the implicit formulation, F acts as an external structural inductive bias encoding prior knowledge that does not need to be learned. This is analogous to an informative Bayesian prior: it concentrates the posterior more rapidly, requiring fewer

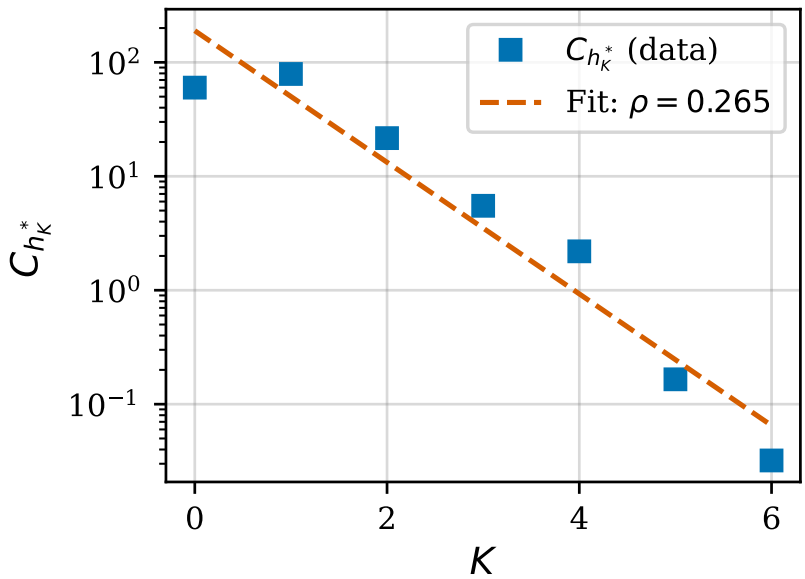

Figure 3: Barron norm $C_{h_K^*}$ of the HAM residual as a function of truncation order K. The geometric decay with rate $\rho_{\text{Barron}} \approx 0.26$ matches the supremum-norm decay, verifying the spectral regularity condition assumed in Proposition 11.

observations. In our framework, the "prior" is the structure of F, and the concentration manifests as reduction of the Rademacher complexity.

The nominal hypothesis class $\mathcal{H}_B$ is the same in both formulations. What changes is the *effective hypothesis space*: since $B_{\text{imp}} < B_{\text{exp}}$, the implicit model restricts the search to a smaller region of parameter space without reducing expressive capacity.

12 Discussion and Future Directions

12.1 What this paper proves

Theorem 6 establishes that the implicit model achieves a tighter generalization bound under the gain condition (13). The experiments confirm this in three regimes: $\beta = 1$ (Experiment 1), HAM residual (Experiment 3), and the null case (Experiment 4). Experiment 2 confirms that large β can negate the advantage.

12.2 Limitations

The paper does not resolve when Assumption 3 holds in general, nor does it explain at a deeper level why certain activation functions are suited to specific dynamics. The HAM proposition (Proposition 11) is conditional on a spectral regularity assumption that remains to be verified rigorously for specific operator classes.

12.3 Toward autosimilar function spaces

Both open questions point toward a topological framework based on autosimilar function spaces, which we outline briefly. The HAM constructs the solution as the limit of an orbit $\{v_k = \Gamma_\hbar^k[u_0]\}$ under a contractive operator, where u_0 is the homogeneous solution of the auxiliary operator $\mathcal{L}$. This orbit is an iterated function system (IFS) in the sense of Hutchinson [15], and the solution u^* is its attractor—not an external target but an intrinsic element of the generated space. The canonical neural architectures (SIREN, RBF, Chebyshev, wavelet networks) are the neural counterparts of the canonical HAM operators. The development and proof of this connection constitute the subject of a companion paper [19].

13 Conclusions

We have established that implicit neural models with structural inductive bias achieve tighter generalization bounds than explicit models. The essential mechanism is target transformation: F changes the learning target from f^* to a simpler h^*, reducing intrinsic approximation complexity. This is fundamentally different from PINN-style regularization.

Two instantiations were identified and validated experimentally: physics-informed variable transformations ($\beta = 1$, $3\times$ advantage), and HAM decompositions ($7\times$ advantage, exponential decay with $\rho \approx 0.45$). The counterexample $\mathcal{S} = 1$ confirms no spurious advantage.

The practical criterion is: compute $\mathcal{S}(F)$ and verify the gain condition. The deeper question—why certain bases are intrinsically adapted to certain dynamics—requires the autosimilar function space framework of the companion paper.

Declaration of competing interest

The author declares no competing interests.

Data availability

Code and data for reproducing all experiments are available at `https://github.com/rodrigo1964-ai/structural-inductive-bias`.

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